A compaction correction for the paleomagnetism of the Nanaimo Group sedimentary rocks: Implications for the Baja British Columbia hypothesis

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[1] A paleomagnetic and magnetic anisotropy study of the Late Cretaceous Northumberland formation on Hornby Island, British Columbia, was conducted to determine if burial compaction could have caused its anomalously shallow inclinations. The shallow Nanaimo Group inclinations have been used to support the Baja British Columbia (Baja BC) model of continental dynamics in which superterranes were transported thousands of kilometers northward along the active North American continental margin in the Cretaceous. A mean of the site means from the Northumberland formation is $D = 5.8°$, $I = 50.9°$, $\alpha_{95} = 7.9°$ ($N = 8$). Anisotropy of magnetic susceptibility (AMS) and anisotropy of anhysteretic remanence (AAR) were measured to identify andcorrect for any inclination shallowing caused by depositional/compactional processes. Both concretion and fine-grained rock samples had typical depositional/compactional AMS and AAR fabrics with bedding perpendicular minimum principal axes. A revised correction equation was used to account for the effects of either triaxial magnetic fabrics or small angular deviations of principal axes from the bedding plane. The average inclination was increased by 9.6° after the compaction correction ($D = 6.1°$, $I = 60.5°$, $\alpha_{95} = 7.1°$, $N = 8$), and the mean inclination of concretion and fine-grained rock sites was significantly steeper than *Ward et al.*'s [1997] result ($I = 42.5°$) placing Hornby Island and the Insular superterrane at a paleolatitude of 41°N in the Late Cretaceous. This paleolatitude indicates that Baja BC did translate northward since the Late Cretaceous by about 1600 km but not by the 3500 km previously indicated. *INDEX TERMS:* 1518 Geomagnetism and Paleomagnetism: Magnetic fabrics and anisotropy; 1525 Geomagnetism and Paleomagnetism: Paleomagnetism applied to tectonics (regional, global); 1527 Geomagnetism and Paleomagnetism: Paleomagnetism applied to geologic processes; 8105 Tectonophysics: Continental margins and sedimentary basins (1212); *KEYWORDS:* paleomagnetism, British Columbia, compaction correction

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1. Introduction

[2] The western Cordillera of North America is composed of exotic terranes that were formed independently and accreted to stable North America at various times during the Mesozoic and Cenozoic. The Insular superterrane, Coast Mountains orogen, and the Intermontane superterrane are three major crustal elements of the northwestern Cordillera [*Cowan et al.*, 1997] (Figure 1) that are situated parallel to the coastline from west to east. While the allochtonous nature of these terranes is agreed upon, there has been a continuing debate about the amplitude of their latitudinal translation. The controversy about the translation centers on the Insular superterrane and Coast Mountains orogen from which paleomagnetic data have suggested >3000 km of displacement since 90 Ma [*Beck*, 1976; *Housen et al.*, 2003; *Irving*, 1985; *Irving et al.*, 1996; *Umhoefer*, 1987; *Ward et al.*, 1997; *Wynne et al.*, 1995]. The Coast Mountains orogen, the boundary zone between the Insular and Intermontane superterranes, was sutured to the Insular superterrane at around 100–90 Ma [*Cowan*, 1994], and the Insular superterrane and Coast Mountains orogen have shared the same displacement history since 90 Ma. Paleomagnetic data have also shown ~1500 km translation of the Intermontane superterrane since the Late Cretaceous [*Irving et al.*, 1995; *Marquis and Globerman*, 1988; *Wynne et al.*, 1998]. *Irving* [1985] has coined the name "Baja British Columbia (Baja BC) hypothesis" to indicate that on the basis of paleomagnetic data a large part of the northwestern coast of the United States and Canada lay near present-day Baja California in Late Cretaceous time. This placement would imply coast-parallel movement of tectonostratigraphic terranes since the Cretaceous. However, the geo-

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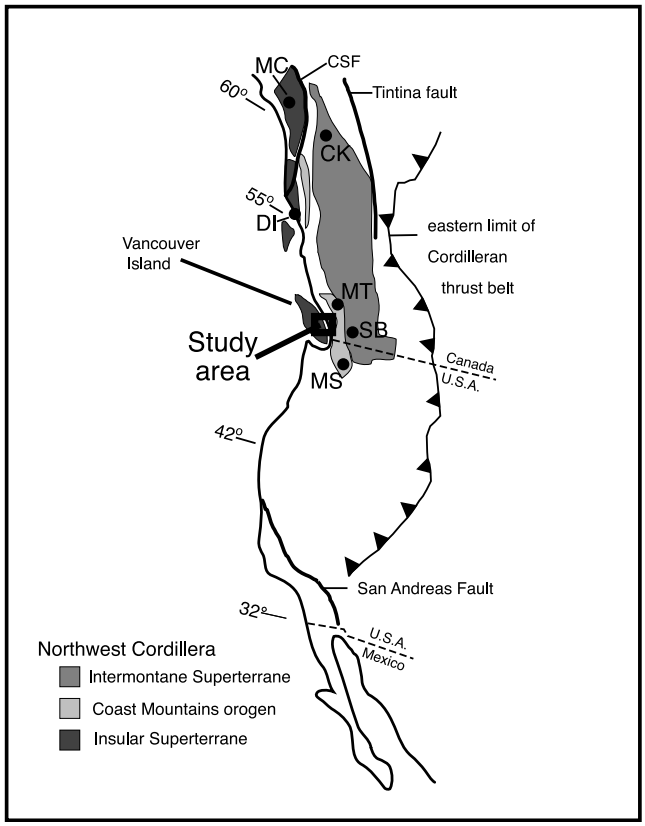


Figure 1. Northwestern Cordillera showing the Insular superterrane, the Coast Mountains orogen, and the Intermontane superterrane (modified from *Cowan et al.* [1997]; reprinted by permission of the *American Journal of Science*). The study area, Hornby Island is located east of Vancouver Island. Circles indicate the localities that yielded paleomagnetic results. MS, Mount Stuart batholith; MT, Mount Tatlow; SB, Spences Bridge; DI, Duke Island; CK, Carmacks Group; MC, MacColl Ridge. Dark lines represent major faults. CSF, Chatham Strait fault. See color version of this figure in the HTML.

logic evidence has suggested that the Insular and Intermontane superterranes were formed near their present position and their northward displacement was <1000 km [*Monger and Price*, 1996]. The absence of megashear faults to accommodate the translation of Baja BC and space problems were used to argue against the Baja BC hypothesis [*Monger and Price*, 1996; *Price and Carmichael*, 1986]. *Haggart and Carter* [1994] argued that 100–145 Ma mollusk fossils and radiolarian microfossils from the Insular superterrane were characteristic of a temperate affinity suggesting northern latitudes for the Insular superterrane in the Late Cretaceous.

[3] *Cowan et al.* [1997] suggested four critical tests of the two mutually contradictory hypotheses for the paleogeography of western North America in the Late Cretaceous. Two of the tests involved provenance studies of detrital zircons deposited in the Late Cretaceous basins of the Insular superterrane and the Coast Mountains orogen. *Mahoney et al.* [1999] have shown Archean (>2.5 Ga) and early Proterozoic (>2.05 Ga) U-Pb ages for detrital zircons from the Methow basin in the Coast Mountains orogen and the Nanaimo and Queen Charlotte basins in the Insular superterrane, where the sediments were derived from source areas to the east. They argued that the presence of Precambrian age zircons indicates that the basins were located at latitudes greater than 40°N and thus disproves the Baja BC hypothesis. However, *Housen and Beck* [1999] examined the same data set in addition to the zircon age data from *Mustard et al.* [1995] and concluded that the peaks in the age distribution of the detrital zircons from older to younger formations in the Nanaimo Group reflected the northward translation of the basin from ~37°N (80 Ma) to ~43°N (65 Ma) and the corresponding change of the sediment source from cratonic North America.

[4] There has been an ongoing effort to resolve the discrepancy between the geologic and paleomagnetic observations by improving the interpretation of paleomagnetic inclination data that are the basis for the paleolatitude determinations [*Butler et al.*, 2001a; *Dickinson and Butler*, 1998]. Anomalously shallow inclinations can result from unreliable cratonic reference paleopoles, unrecognized tectonic tilting of plutonic igneous rocks, and compaction of sedimentary rocks [*Dickinson and Butler*, 1998]. The paleomagnetic data from the Mount Stuart batholith [*Beck et al.*, 1981] exemplify the case in which the tilting of the batholith could be the cause of the anomalously shallow inclinations [*Butler et al.*, 1989]. The original paleomagnetic data from the Mount Stuart batholith yielded a paleolatitude of 27° ± 4.0°N at 93–82 Ma that indicated a displacement of 3000 km for the Coast Mountain orogen [*Beck et al.*, 1981]. *Butler et al.* [1989] argued that removing a 35° tilt to the southwest would bring the shallow inclination to the expected inclination. *Ague and Brandon* [1996] examined the possible tilting of the strata by using Al-in-hornblende geobarometry (AH) and found evidence for only 7° tilt to the southeast. *Housen et al.* [2003] revisited the Mount Stuart batholith and their paleomagnetic data reconfirm the previous results reported by *Beck et al.* [1981]. *Housen et al.*'s [2003] Ar-Ar geochronology for hornblende and biotite suggests that the remanence carried by the single domain magnetite in the southern part of the batholith was acquired at 91 Ma within 0.5–1.0 m.y. of the closure of Al in hornblende. Their direction after the correction for the tilt indicated by AH barometry still yields a paleolatitude of 31.3° ± 3.5°N at 91 Ma.

[5] The 110 m.y. old Duke Island ultramafic complex also yielded a shallow inclination, after a tilt correction based on the regional trend of cumulate layers, indicating 2970 ± 1360 km of translation for the Insular superterrane [*Bogue et al.*, 1995]. However, the paleomagnetic directions after a tilt correction using cumulate layers at the site level are reported to yield a scattered result [*Butler et al.*, 2001b]. The paleomagnetic study of the Silverquick conglomerate and the Powell Creek volcanics from the Mount Tatlow area of British Columbia [*Wynne et al.*, 1995] is considered to be one of the most robust pieces of data supporting the Baja BC hypothesis. The paleomagnetic data from 98–74 Ma sedimentary and volcanic rocks passed a fold test and the directions from each lithology were in agreement, suggesting 2970 ± 450 km of northward translation for the Coast Mountains orogen since the Cretaceous.

[6] The Nanaimo group sedimentary rocks from the Insular superterrane have attracted paleomagnetic researchers

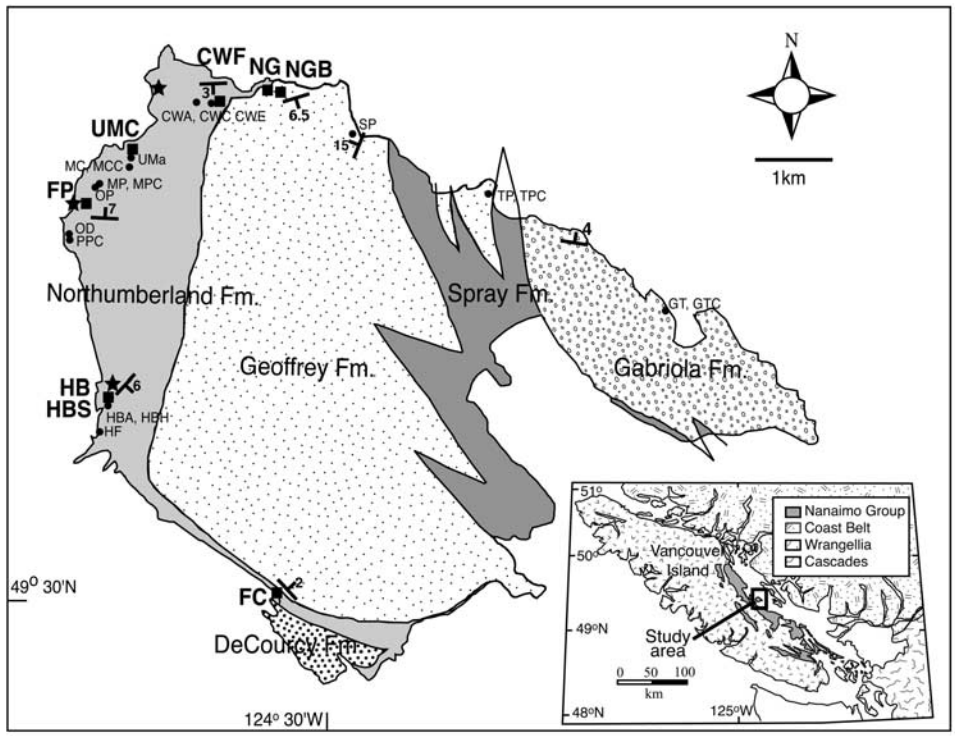


Figure 2. Geologic map of Hornby Island (modified from *Katnick and Mustard* [2001] with permission from British Columbia Ministry of Energy and Mines) and paleomagnetic sampling locations. From the sampling sites from three field seasons, only the sampling sites with available GPS coordinates (27 sites out of 36 sites) were marked in this map. Squares represent sites which yielded the ChRM used for the calculation of the mean direction from the Northumberland formation. Stars indicate the sampling locations from *Ward et al.* [1997]. Inset map shows the extent of the Nanaimo Group and the location of Hornby Island. See color version of this figure in the HTML.

because the well-bedded sedimentary rocks provide clear evidence of paleohorizontal [*Enkin et al.*, 2001; *Ward et al.*, 1997]. The paleomagnetic data from the Nanaimo Group reported by *Ward et al.* [1997] were collected from silty mudstone and carbonate concretion samples and showed much shallower paleomagnetic inclinations ($I = 42.5°$ from Hornby and Texada islands) than the inclinations calculated from the Late Cretaceous reference paleopole ($I = 70.0°$). This result places the Insular superterrane, on which Nanaimo Group sedimentary rocks were deposited, at a very low paleolatitude (24.6°N), lower than any previous results.

[7] However, fine-grained sedimentary rocks can suffer from paleomagnetic inclination shallowing during the post-depositional process of burial compaction [*Anson and Kodama*, 1987; *Deamer and Kodama*, 1990; *Sun and Kodama*, 1992]. Pigeon Point Formation turbidites [*Kodama and Davi*, 1995], the Cretaceous Ladd Formation, and the Point Loma Formation turbidites [*Tan and Kodama*, 1998] were shown to have 10–17° inclination shallowing, probably due to burial compaction. Since clay content apparently plays an important role in causing anomalously shallow inclinations during burial compaction [*Anson and Kodama*, 1987; *Deamer and Kodama*, 1990; *Jackson et al.*, 1991; *Sun and Kodama*, 1992; *Tan and Kodama*, 1998], there is a possibility that compactional processes in the sedimentary rocks have caused the lower inclination observed in the Nanaimo Group sedimentary rocks by *Ward et al.* [1997]. The steeper magnetic inclination from sandstones of the Nanaimo Group ($I = 55.2°$) reported by *Enkin et al.* [2001] also supports the possibility of compaction-caused inclination shallowing in Ward et al.'s data.

[8] Besides the advantage of providing an unambiguous measure of paleohorizonal, the age of the upper Nanaimo Group sedimentary rocks provides an important constraint in constructing the time-space evolution of the Insular superterrane. The paleomagnetic data from the Mount Stuart batholith and the Mount Tatlow region suggest that the Insular superterrane was located near 30°N at ~90 Ma. The translation of the Insular superterrane should have been completed by 50 Ma [*Fox and Beck*, 1985; *Irving and Brandon*, 1990; *Irving et al.*, 1996]. The Nanaimo Group was deposited during the period of proposed translation [*Mustard and Monger*, 1994] and thus are expected to record the latitudinal change in the sediments.

[9] In this study, the paleomagnetic data were collected from the upper Nanaimo Group sedimentary rocks in Hornby Island and rock magnetic studies were carried out to identify and correct any possible inclination shallowing by burial compaction. The new paleolatitude obtained from the corrected inclination places the Nanaimo Group at a position consistent with the displacement trajectories of *Housen and Beck* [1999] and *Kodama and Ward* [2001] and provides a new constraint for the paleogeographic

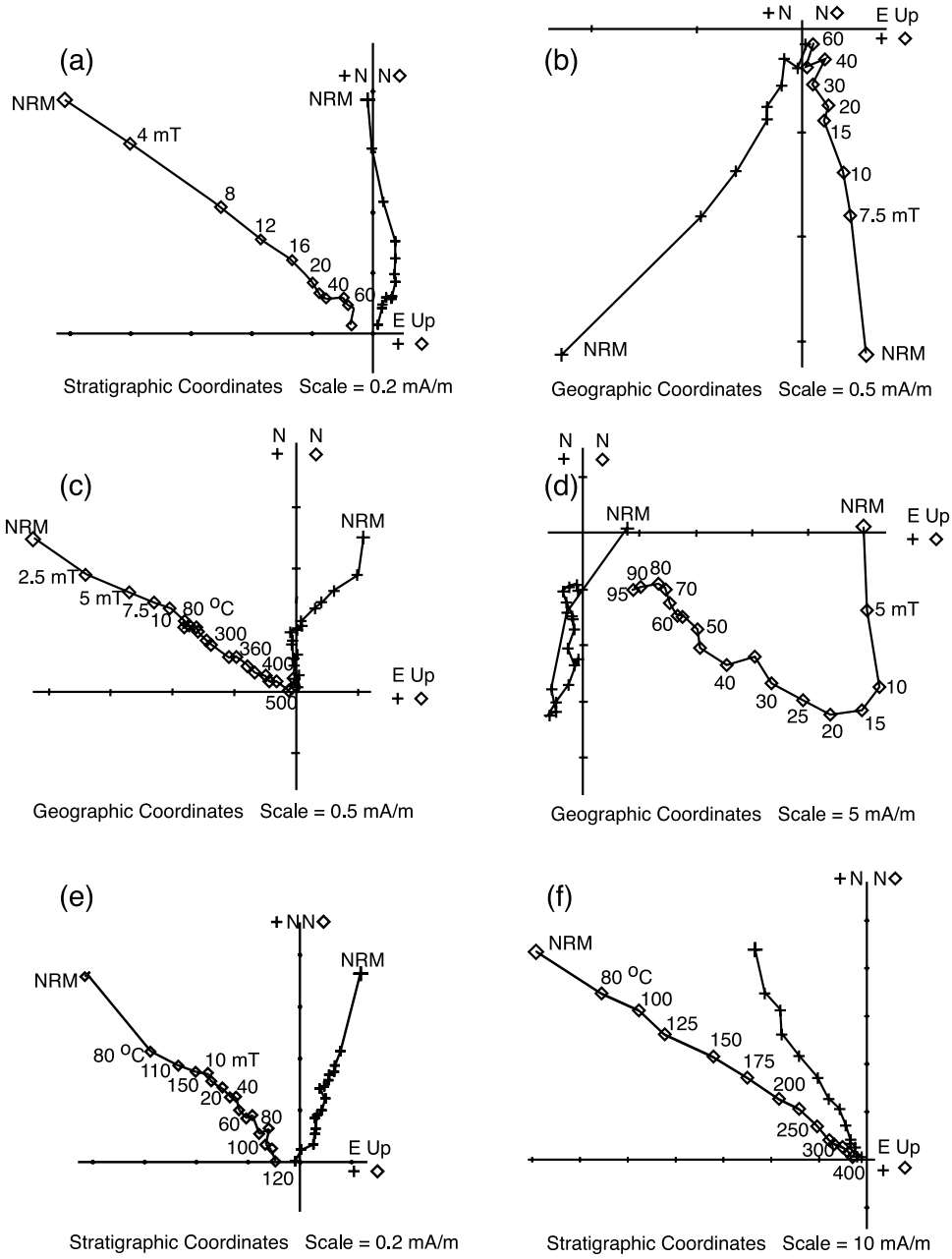


Figure 3. Vector endpoint diagrams showing demagnetization behavior of representative samples from (a) NGB mudstone, (b) HBS mudstone, (c) UMC concretion, (d) HB concretion, (e) NG turbidite, and (f) CWF sandstone. See color version of this figure in the HTML.

reconstruction of the Insular superterrane in the Late Cretaceous.

2. Geology

[10] The Nanaimo Basin, in which the Nanaimo Group sedimentary rocks are deposited, is bounded by the Wrangellia terrane to the west, the Coastal Plutonic Belt to the east, and North Cascades to the southeast (Figure 2). The Nanaimo Group comprises a sequence of marine sedimentary rocks deposited in the Late Cretaceous in a marine shelf or submarine fan environment and is mainly exposed in the eastern part of Vancouver Island and in the islands of the Straits of Georgia [*Mustard et al.*, 1999].

[11] The upper part of the Nanaimo Group is exposed on Hornby Island. The De Courcy Formation, the oldest formation exposed on Hornby Island, covers the southwestern tip of the island. The Northumberland formation, the next oldest formation, covers the area along the island's western shoreline. The younger formations of the Nanaimo Group conformably lie subparallel to each other to the northeast (Figure 2). The formations are not severely deformed, and have a gentle dip on the average of 8° to the NE and strike roughly NW-SE [*Katnick and Mustard*, 2001]. A detailed sedimentalogical study shows that several sedimentation cycles deposited coarse- to fine-grained sediments, from conglomerate-dominant formations such as the De Courcy, Geoffrey, and Gabriola formations to

Table 1. Site-Mean Data From the Northumberland Formation[a]

			Site Location				Before Compaction Correction				After Compaction Correction				
Sampling Site	Rock Types	S/D	Latitude	Longitude	*n*/*N*	Nc	*D*	*I*	*k*	a_{95}	*D*	*I*	*k*	a_{95}	ΔI
Ngb	shale	73/6.5	124.67704	49.55129	6/6	5	10.6	46.3	28.5	12.8	14.7	54.0	39.6	12.3	7.7
Ng	siltstone, fine sandstone	33/3.5	124.67904	49.55158	19/22	13	11.2	48.3	26.1	8.3	10.4	52.6	20.7	9.3	4.3
Cwf	sandstone	88/3	124.68749	49.55025	8/8	7	346.6	65.4	43.1	8.5	347.7	75.1	58.7	7.9	9.7
Umc	carbonate concretion	0/0	124.70213	49.54413	5/5	5	348	58.8	138.1	6.5	337.9	66.4	77.1	8.8	7.6
Fp	shale	270/5	124.70977	49.53903	7/7	7	11.4	46.4	19.7	14	9.9	55.8	22.0	13.2	9.4
Hbs	shale	0/0	124.70480	49.51922	11/18	10	180.6	−35.2	18.5	11.5	178.6	−55.3	28.3	9.2	20.1
Hb	concretionary bed	43/6	124.70493	49.51901	17/18	12	198.0	−51.4	45.0	6.5	201.8	−65.0	53.3	6.0	13.6
Fc	carbonate concretion	317/2	124.67559	49.49921	9/12	6	8.6	50.9	24.4	10.6	12.2	55.4	31.6	12.1	4.5
	shale (Ngb, Fp, Hbs)				3[b]		7.5	43.2	94.5	12.8	7.8	55.2	281.7	7.4	12.0
	concretion (Umc, Hb, Fc)				3[b]		6.5	53.8	63.4	15.6	5.5	63.4	48.5	17.9	9.6
Mean						8[b]	5.8	50.9	49.8	7.9	6.1	60.5	61.2	7.1	9.6

[a]S/D, strike/dip; *n*/*N*, the number of samples which yield ChRM direction/the number of samples measured; Nc, the number of samples used for the compaction correction; *D*, *I*, declination, inclination; *k*, precision parameter; a_{95}, 95% confidence interval of the mean;. ΔI, the amount of inclination shallowing. NGb, Northumberland Geoffrey site b; Ng, Northumberland and Geoffrey contact; Cwf, Collishaw point site f; Umc, Upper Manning concretion; Fp, ferry phipps point; Hbs, Hornby ferry shale; Hb, Hornby ferry concretion; Fc, Ford cove concretion.
[b]Number of sites.

finer mudstone and sandstone formations such as the Northumberland and Spray formations [*Mustard et al.*, 1999].

[12] A paleocurrent analysis indicates dominant flow directions to the north and to the west suggesting that the sediment source area was from the Coastal Plutonic belt in the east and the North Cascades in the southeast [*Katnick and Mustard*, 2001]. Carbonate concretions are imbedded in the mudstones and sandstones, and are especially abundant in the Northumberland formation. The draping of host layers around concretions was observed in the field and a measurement of difference in the thickness of layers found in a concretion and its host rocks showed the ratio of 1.5. Macrofossils found in the Northumberland formation in Hornby Island suggest an Upper Campanian (~75 Ma) age for the formation, but the ages for the younger formations are not certain and possibly extended into the Maastrichtian [*Ward*, 1978].

3. Sampling

[13] Samples were collected on three sampling trips in the summers of 1998, 1999, and 2001 (Figure 2). Previously collected paleomagnetic samples from Hornby Island are notorious for their hard-to-interpret demagnetization behavior [*Enkin et al.*, 2001]. This was found in our study, too, and necessitated multiple trips to collect enough samples and sites with reasonable quality results.

[14] In the first sampling trip in 1998, 93 samples from 15 sites were collected mainly from the Northumberland and Geoffrey formations. The Northumberland formation is particularly interesting for three reasons. The Northumberland formation is the oldest fine-grained rock formation exposed in the island, and has unaltered ammonite fossils and carbonate concretions. The age of the formation is well constrained (~75 Ma), and it is expected to yield a shallower inclination compared to younger formations if the Insular superterrane was translated as proposed in the Baja BC hypothesis. Unaltered fossils in the Northumberland formation indicate that the sampling area was not heated more than 80°C [*Ward et al.*, 1997]. The paleomagnetic data from concretion samples were assumed to be free from inclination shallowing by compaction, which was used as an argument that the mudstones, which yielded a direction statistically indistinguishable from the direction observed in the concretions, did not suffer from inclination shallowing [*Ward et al.*, 1997]. The paleomagnetic and rock magnetic differences between the mudstones and the carbonate concretions from the Northumberland formation were examined in this study. In 1999, the collection was extended to the younger Gabriola Formation and 103 samples from 11 sites were collected. This collection was intended to test the idea of "transport magnetostratigraphy" [*Housen and Beck*, 1999] and to investigate if the samples from younger formations show steeper inclinations. In the last sampling trip (2001), the Northumberland formation was targeted because some of the sites from this formation collected in 1998 had yielded stable paleomagnetic data. The samples collected in 1998 suggested that mudstone samples had shallower inclinations than carbonate concretion samples, but the number of samples and sites from the 1998 collection was small and resulted in large 95% confidence limits for the site means. This trip gave the opportunity to collect more samples and sites (237 samples from 20 sites) from the Northumberland formation and thus obtain statistically better constrained results.

[15] A total of 433 oriented cores from 36 sites were collected from the three field trips. Core samples were drilled using a gasoline-powered portable drill and the orientation of cores was measured using a Brunton compass. Core samples were brought to the laboratory, cut into 22 mm long cylinders, and paleomagnetic and rock magnetic measurements were carried out on the samples.

4. Paleomagnetic Data

4.1. Paleomagnetic Measurements

[16] Standard paleomagnetic measurements were performed to extract the characteristic remanent magnetization (ChRM) from each sample. Detailed demagnetization for four samples from each site was conducted, two using alternating field (AF) demagnetization and two using thermal demagnetization. Stepwise demagnetization was carried out in more than 10 steps using 2–10 mT increments for AF demagnetization and 20–50°C increments for thermal demagnetization. One or two additional samples were treated using a combination of demagnetization techniques.

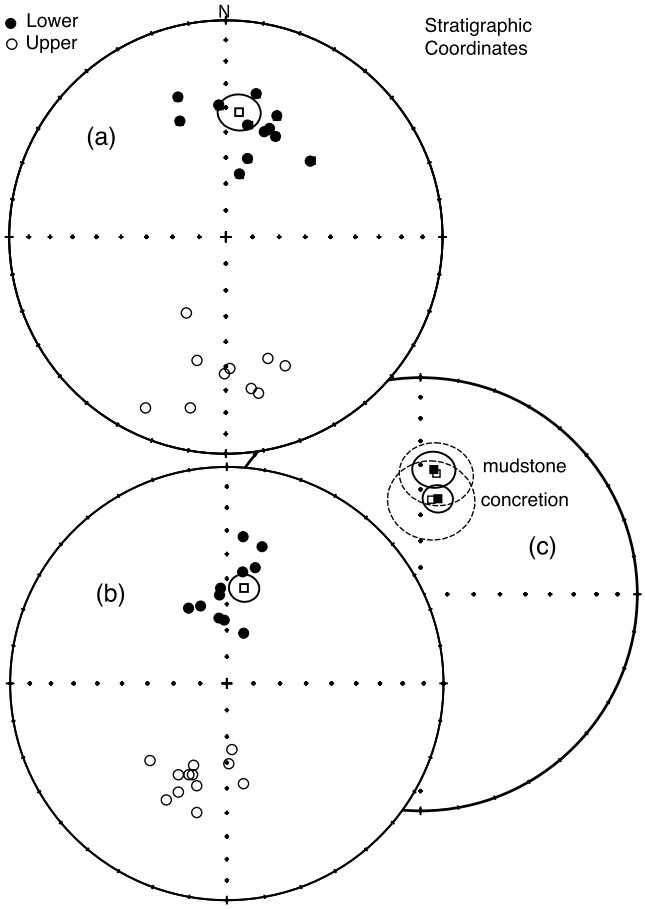


Figure 4. The ChRM directions of (a) mudstone samples from NGB, FP, and HBS sites and their mean direction with α_{95} and (b) carbonate concretion samples from UMC, HB, and FC sites and their mean direction with α_{95}. (c) Comparison of the mean directions from the mudstone and concretion sites. Solid squares and solid line represent the mean directions with the confidence angle calculated at the sample level, while open squares and dashed lines represent the mean directions with the confidence angle calculated at the site level. Note that the difference between mudstone and concretion is statistically significant at 90% confidence level if the means are calculated at the sample level. The difference is undistinguishable at the site level because of large α_{95} from small N (N = 3 for each lithology).

Low-field AF demagnetization up to 10 mT with 2 mT intervals was followed by thermal demagnetization from 80–100°C up to 550–600°C in 12–20 steps. After these pilot measurements, AF demagnetization was selected as the most effective method and applied to the rest of the samples in more than 10 steps. In addition, the use of AF demagnetization avoided the chemical/mineralogical changes often caused by thermal demagnetization. The samples were measured in laboratories equipped with 2G superconducting rock magnetometers housed in magnetically shielded rooms. The nominal field in these rooms was less than 1000 nT. AF demagnetization was performed using the AF demagnetization unit installed in the 2G superconducting rock magnetometer. A thermal demagnetizer designed and built by D. Kent at Lamont Doherty Earth Observatory and a Schonstedt TSD-1 thermal demagnetizer were used for the 1998 and 1999 samples. An ASC Scientific TD-48 SC thermal demagnetizer was used for the thermal demagnetization of the 2001 samples. The demagnetization results were analyzed using vector endpoint diagrams [*Zijderveld*, 1967], and the characteristic remanent direction was calculated using principal component analysis [*Kirschvink*, 1980] and great circle analysis [*Halls*, 1976]. Fisher statistics was used to calculate site-mean directions [*Fisher*, 1953].

4.2. Paleomagnetic Direction

[17] In many cases, samples carried a weak magnetization and did not show stable demagnetization behavior in vector endpoint diagrams. Both normal and reversed polarities were observed in the NRM directions. Many fine-grained rocks had low coercivities and the demagnetized intensity dropped below 10% of the NRM by an AF demagnetization field of 20 mT. Most of the normal polarity samples, including the concretion samples, showed a big drop (~50%) in intensity by AF demagnetization at 10 mT (Figure 3). Samples with reversed polarity often showed an increase in intensity by AF demagnetization at 10 mT indicating the removal of a normal polarity overprint (Figure 3d).

[18] Demagnetization of the pilot samples did not show much difference in their behavior between thermal and AF demagnetization. Even though some thermally demagnetized samples showed a greater decrease in their intensity than AF demagnetized samples, the directions became unstable when they reached about 10% of the NRM. The sandstone site (CWF) and two of the mudstone sites (NGB, HBS) showed lower MDFs than other mudstone and concretion sites, but their ChRMs were still extracted at peak AF fields greater than 10 mT of AF demagnetization. The highest thermal demagnetization temperatures at which the ChRMs were isolated ranged from 225°C to 560°C, and are typically about 400–500°C (Figures 3c and 3f).

[19] On the basis of the directional changes and the decay of intensity during the demagnetization, either principal component analysis or great circle analysis with constraints was applied. Principal component analysis was applied when more than three demagnetization steps in the vector endpoint diagram decayed linearly to the origin with a corresponding substantial decay in intensity (more than 30%). Great circle analysis was applied when stereonet plots showed that the demagnetized directions traveled along a great circle for an angular distance greater than 15°. Only the ChRM directions with maximum angular deviations (MADs) less than 10° (15° for the great circle fit) were used for the calculation of the site means. Site means with α_{95} larger than 15° were rejected. This led to discarding the sites based predominantly on great circle analysis. Only directional data obtained by principal component analysis (PCA) were used in calculating the site means [*Fisher*, 1953] and only 8 sites out of 36 sites were accepted based on the size of their α_{95}. The site means are all obtained from the Northumberland formation, including three sites derived from carbonate concretions, three sites from mudstones, one site of fine-grained sand- and silt-sized turbidites, and one site of fine-grained sandstones (Table 1).

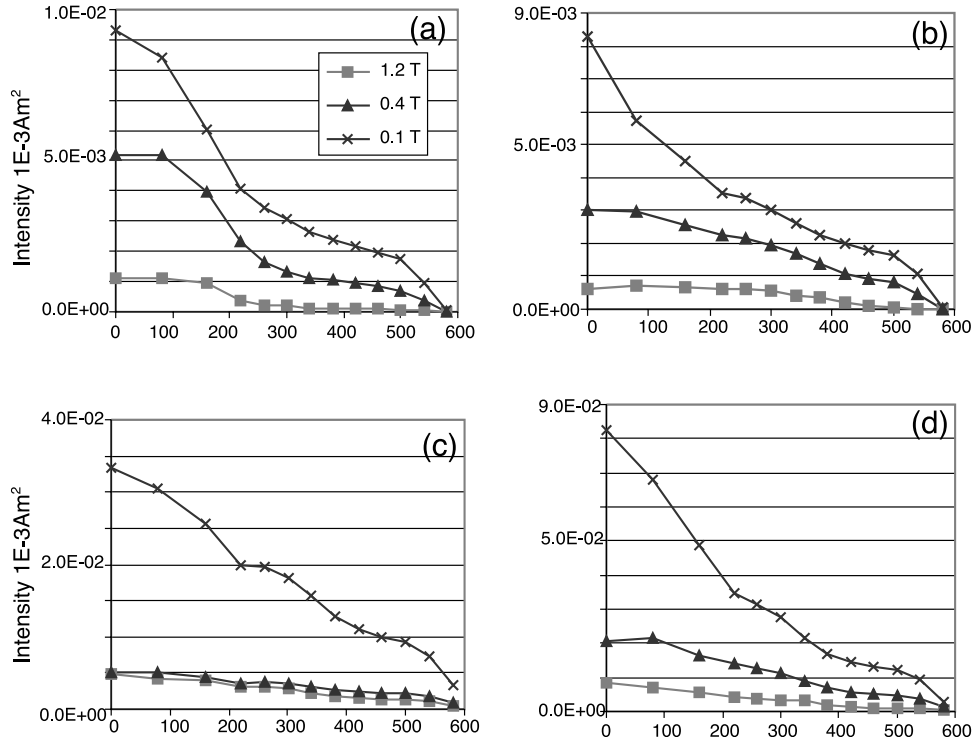


Figure 5. Thermal demagnetization of a three component isothermal remanent magnetization (IRM). The 1.2, 0.4, and 0.1 T IRM fields were applied to orthogonal axes of the samples before the thermal demagnetization. Samples are from (a) NG turbidite, (b) UMC concretion, (c) HBS shale, and (d) HB concretion.

The pairs of sites collected in two different sampling years but from the same location were combined to calculate the site mean for the site.

[20] The mean directions of the mudstone samples (from sites NGB, FP, and HBS) and concretion samples (from sites UMC, HB, and FC) are compared in Figure 4. The mean direction of all the mudstone samples is shallower than the concretion samples' mean by 11.4°. The mean of the mudstone samples averaged at the sample level is $D = 6.1°$, $I = 41.7°$, $\alpha_{95} = 7.4°$ ($n = 22$) and the mean of the concretion samples is $D = 10.2°$, $I = 53.1°$, $\alpha_{95} = 5.4°$ ($n = 23$). These two mean directions are statistically different at the 90% confidence level. However, the mean directions at the site level for each lithology are statistically indistinguishable at 95% confidence level ($N = 3$ each for mudstone and concretion sites). The mean of the all sites for the Northumberland formation is $D = 5.8°$, $I = 50.9°$, $\alpha_{95} = 7.9°$ ($N = 8$).

5. Rock Magnetic Data

[21] Partial anhysteretic remanences (pARM) were applied to one sample from each site after AF demagnetization. A DC field of 0.1 mT in a 10 mT AF window from 0 to 100 mT was used to impart the pARMs. The pARM spectra of most samples show a peak in coercivity around 30–60 mT. The remainder of the samples shows another coercivity peak at 0–10 mT indicating an enrichment of low coercivity grains (e.g., samples from HBS site). An interesting observation is that the seven samples with the strongest pARM intensities come from the sites with well-defined paleomagnetic directions during demagnetization (HB, FC, HBS, NG, and NGC). This supports the observation that the unstable demagnetization behavior is partly due to the weak NRM intensity of the samples.

[22] The thermal demagnetization of the isothermal remanent magnetization (IRM) showed continuous decay in intensity (Figure 5). The IRM component imparted at 0.1 T carried most of the remanence while the components imparted at 0.4 and 1.2 T showed similar behavior to the 0.1 T component in their decay during thermal demagnetization. One component of magnetization is removed below 200°C. The higher unblocking temperature components occur around 400°C and at 580°C. These unblocking temperature components were also found in the thermal demagnetization of NRM.

5.1. Magnetic Fabric

[23] Anisotropy of magnetic susceptibility (AMS) was measured for all samples, before the paleomagnetic measurements, using an Agico KLY-3S Kappabridge. The coercivity windows used for anisotropy of anhysteretic remanence (AAR) measurements were determined from the AF demagnetization behavior and pARM coercivity spectra. The AF windows used were selected to target the magnetic grains carrying the characteristic remanence (ChRM). In order to determine the remanence anisotropy ellipsoid, pARMs were applied in nine different orientations, and AF demagnetizations were carried out between the different orientations used for pARM application [*McCabe et al.*, 1985]. The orientation and the shape of the AAR ellipsoid and directions of the principal axes were calculated following the techniques of *McCabe et al.* [1985]

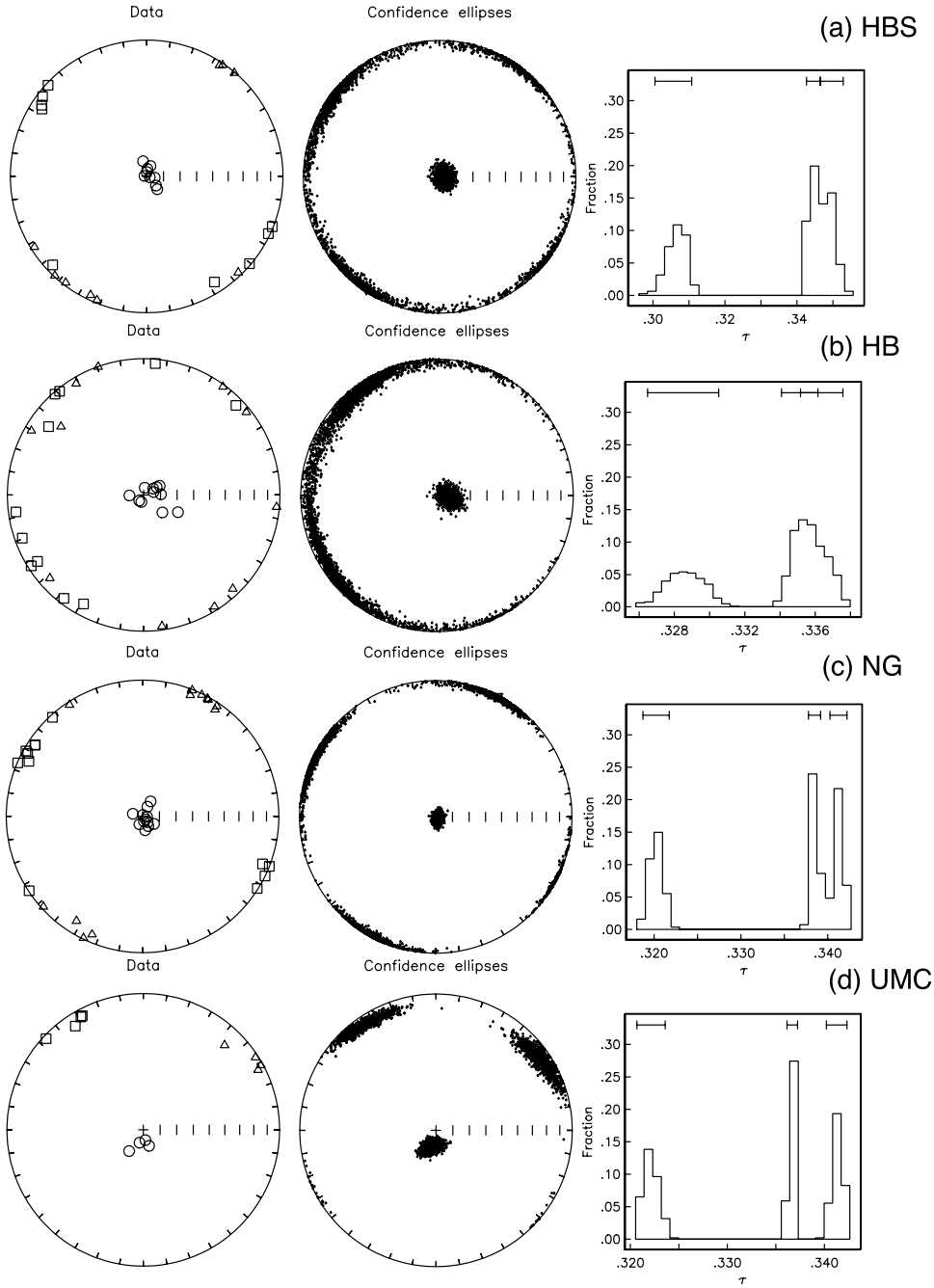


Figure 6. Lower hemisphere equal-area projections of anisotropy of magnetic susceptibility (AMS) data, bootstrap eigenvectors, and histograms of the bootstrapped principal axes (τ): (a) HBS, (b) HB, (c) NG, and (d) UMC. Maximum principal axes are shown by squares, intermediate axes by triangles, and minimum axes by circles.

and compared with the AMS results using bootstrap statistics [*Constable and Tauxe*, 1990].

[24] Only the data for samples that have both AMS and AAR data are presented for direct comparison between the two magnetic fabrics. Bootstrapped eigenvectors and the histogram of the eigenparameters generated by the bootstrap programs [*Tauxe*, 1998] present the orientation and shape anisotropy of the AMS and AAR data. Most of the AMS results show a depositional/compactional fabric with minimum axes clustered near the perpendicular to bedding (Figure 6).

[25] AAR data from all the sites show minimum principal axes clustered perpendicular to bedding except for two sites (UMC and NG). These two sites still show horizontal maximum principal axes. Two mudstone sites (HBS and NGB) and two concretion sites (FC and HB) show depositional/compactional AAR fabric (Figure 7). A mudstone site, FP and the CWF sandstone sites show triaxial AAR fabrics. The AAR fabric of the turbidite site (NG) shows a triaxial fabric with the minimum axes tilted toward the southeast (Figure 7c). This is different from its AMS results that show a typical depositional/compactional fabric com-

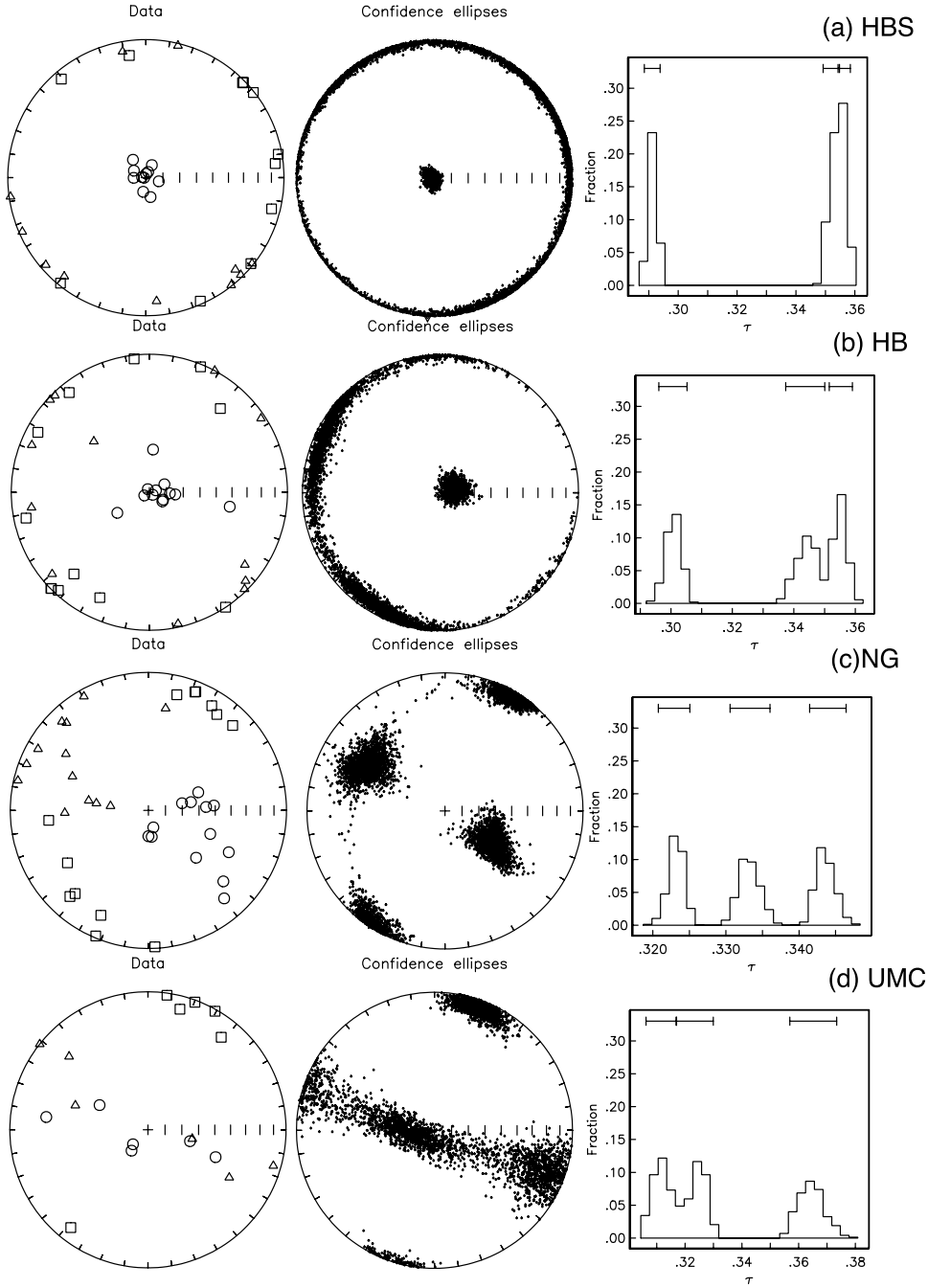


Figure 7. Lower hemisphere equal-area projections of anisotropy of anhysteretic remanence (AAR) data, bootstrap eigenvectors, and histograms of the bootstrapped principal axes (τ): (a) HBS, (b) HB, (c) NG, and (d) UMC. Maximum principal axes are shown by squares, intermediate axes by triangles, and minimum axes by circles.

mon in sedimentary rocks (Figure 6c). A concretion site, UMC shows a prolate AAR fabric. The maximum axes are horizontal and clustered in the NNE direction, while the minimum and intermediate axes are scattered along the great circle trending east to west (Figure 7d).

5.2. The *a* Factor

[26] The anisotropy of individual magnetic grains, the *a* factor, was estimated by three different methods. In the first method, compaction experiments [*Kodama*, 1997] were used, but the Hornby Island samples did not give easily interpretable compaction results. Therefore the *a* factor could not be determined in this way. The epoxy drying method with magnetic extracts was used to determine the *a* factor and the average *a* factor obtained from this method was used for the compaction correction. Shape ratios of magnetic grains observed under the microscope were also used to determine the *a* factor, primarily to check the value obtained from the epoxy drying method.

5.2.1. Epoxy Drying Method

[27] Magnetic minerals were separated from the slurry made from disaggregating samples using techniques

described by *Hounslow and Maher* [1998]. The magnetic extract was stirred into a 11 cm^3 sample cylinder filled with clear quick-drying epoxy. The sample was allowed to dry and hardened in magnetic fields so that the magnetic grains would align their magnetic easy axes parallel to the magnetic field. The ratio between the magnetic moments acquired along the axes perpendicular and parallel to the applied magnetic field direction represents the average *a* factor of the magnetic particles in the sample. Only a small amount of magnetic grains (less than 0.01 g) were dispersed in the epoxy to minimize the interaction of the magnetic grains and the formation of needle-shaped clumps of the magnetic grains. Static magnetic fields of 50 mT were shown to completely align magnetic grains in other epoxy-drying experiments [*Jackson et al.*, 1991; *Kodama*, 1997; *Tan and Kodama*, 1998]. Magnetic fields larger than 50 mT will impart a strong IRM to the magnetic grains in the epoxy, and thus DC magnetic fields of 50 mT were used for this study.

[28] Two ARM windows of 10–30 mT and 20–60 mT were applied to measure the *a* factor (Figure 8). An AAR window of 10–30 mT was used for the epoxy samples containing magnetic extracts from the fine-grained rock sites, from which the ChRMs were isolated at lower coercivities. An AAR window of 20–60 mT was used for the epoxy samples containing magnetic extracts from the concretion sites, which showed higher coercivities to isolate their ChRMs. The mean *a* factor from fine-grained rocks, using a pARM coercivity window of 10–30 mT, is 2.34, and the mean *a* factor from concretions, using a pARM coercivity window of 20–60 mT, is 2.36. Since there was not significant difference between the two *a* factors, the mean of *a* factors from both fine-grained rocks and concretions, 2.35 ± 0.55 (1 SD), was used for the compaction correction.

5.2.2. Direct Observation

[29] Two approaches were used for the direct observation of the magnetic grains. First, the epoxy-mounted samples from the magnetic extraction experiments were polished and observed under a reflected-light petrographic microscope. A magnification of 1000 was used and the long and short axes of grains larger than 1 μm were digitally measured with an image analysis program. Second, polished paleomagnetic core samples and magnetic extracts placed on carbon disks were observed under a JEOL JSM-6300F field emission scanning electron microscope (SEM), and energy-dispersive X-ray spectra (EDS) were obtained from point analysis of the targeted grains. Once dimensional ratios of the magnetic grains are known, the *a* factor can be estimated following the method of *Stephenson et al.* [1986] (Appendix A).

[30] The epoxy sample made from an HB concretion sample had the greatest abundance of grains and was used for the shape ratio method of estimating the *a* factor. Direct observation of 101 magnetic grains between 1 and 6 μm in size yielded an average grain shape ratio of 1.41 ± 0.38 for the site, and a self-demagnetizing factor of N_n = 0.38 is derived based on the work by *Stacey and Banerjee* [1974]. The correlation between the principal axes of AMS and AAR, using equation (A1), yielded p_0 = 0.1744 with an R^2 = 0.73. With p_0 = 0.1744 and N_n = 0.38, the intrinsic susceptibility of χ_i = 96 is obtained, finally yielding a calculated *a* factor of a_r = 2.1. This estimation of the *a* factor, a_r, is similar to the *a* factor obtained from direct measurement in the epoxy-drying experiment.

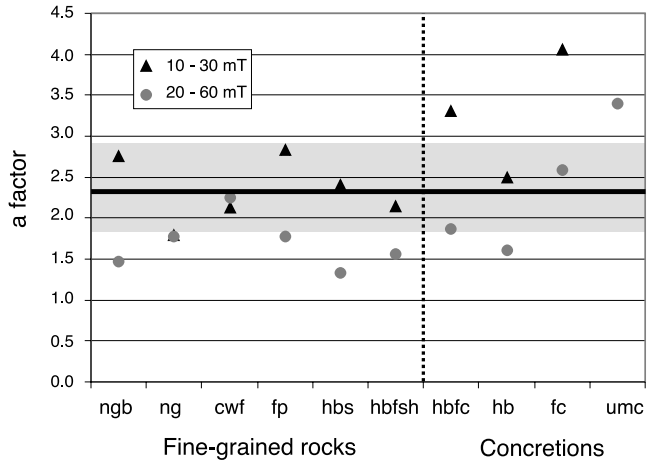


Figure 8. The *a* factors measured from magnetic extracts mounted in clear epoxy and dried in a static magnetic field of 50 mT. The legend shows the range of the applied pARM window to obtain the *a* factors. The values on the left side of the dashed line are from the fine-grained rocks, and those on the right side are from the concretion samples. Note that the *a* factors from the lower ARM window show larger *a* factors, and those from the concretion samples have a larger variation between samples. The mean *a* factor used for the correction and the standard deviation are shown as a solid line in the shaded area. See color version of this figure in the HTML.

[31] Scanning electron microscope (SEM) observation of the polished rock samples and the magnetic extract showed detrital grains with sharp edges indicating a short distance of transport from the source area (Figure 9). The size of the detrital grains observed under the SEM ranged from 1–2 μm to ~10 μm with the physical grain shape ratios (length:width) ranging from 1.2 to 1.8. Energy-dispersive X-ray spectra (EDS) analysis showed that the detrital grains were Fe oxides and Fe-Ti oxides. Unblocking of NRM and IRM below 580°C in thermal demagnetization suggests the small magnetic grains with Fe peaks are probably magnetite. The grains with Fe and Ti peaks are probably responsible for unblocking temperature around 400°C. However, euhedral grains with Fe and S peaks were also found in the polished core samples as well as in the magnetic extracts. Nonferromagnetic pyrite framboids commonly found in fine-grained rocks and concretions are probably diagenetic products in the organic-rich sedimentary rocks.

6. Compaction Correction

6.1. Revised Inclination Correction Equation

[32] Previous inclination correction studies [*Jackson et al.*, 1991; *Kodama and Davi*, 1995; *Tan and Kodama*, 2003] assume an oblate DRM anisotropy ellipsoid with minimum axes perpendicular to the bedding plane and the magnetic direction confined to the *XZ* plane. We also assume that the DRM is a product of the DRM anisotropy tensor and the original magnetic field

$$\mathrm{DRM} = \mathbf{k}_D\mathbf{H}, \quad (1)$$

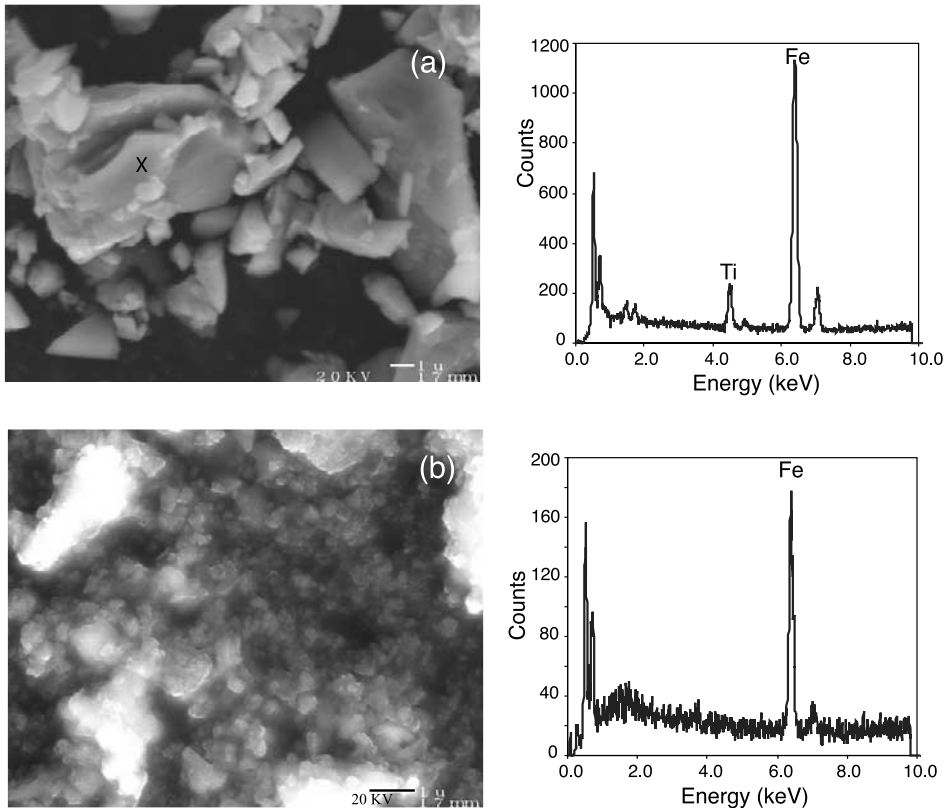


Figure 9. Scanning electron micrographs of magnetic extracts on carbon disk: (a) HB concretion and (b) NGB shale. EDS analysis on the marked grain (Figure 9a) showed Fe and Ti peaks. The shown area in Figure 9b featured a major Fe peak.

and that the best estimate of the DRM anisotropy tensor can be obtained from the ARM anisotropy tensor. However, in our data the AAR anisotropy ellipsoid is not always exactly parallel to the bedding plane and not always oblate. Therefore, in our revised correction equation the DRM anisotropy tensor is allowed to take any orientation and shape, and the original magnetic field direction is restored by the product of the DRM vector and the inverse of the DRM anisotropy tensor (see Appendix B).

[33] In this approach, the corrected inclination is no longer derived from a simple equation. In the revised correction, four steps are followed: (1) the ARM anisotropy tensor and the individual particle anisotropy factor, *a*, are determined from measurement, (2) the principal axes of the DRM anisotropy tensor are calculated from the ARM anisotropy tensor and the *a* factor by equation (B4), (3) the orientation of the DRM anisotropy tensor is determined from the orientation of the ARM anisotropy tensor, (4) the inverse DRM anisotropy tensor is multiplied by the ChRM direction to yield the original magnetic field vector (**H**) which is parallel to the corrected remanence direction. Besides the *a* factor, the eigenvectors of the ARM anisotropy tensor and the three components of ChRM (DRM) now appear in the final equation

$$\left(H_x, H_y, H_z\right) = \mathbf{k_D^{-1}}\left(\mathrm{ChRM}_x, \mathrm{ChRM}_y, \mathrm{ChRM}_z\right). \qquad (2)$$

[34] This revised method allows the orientation of the anisotropy tensor to be fully reflected in the inclination correction of the ChRM. With the revised correction equation (1), a site with its individual samples' maximum and intermediate ARM anisotropy principal axes oriented exactly in the bedding plane will yield a bigger compaction correction compared to a site with its individual samples' maximum and intermediate ARM anisotropy principal axes not exactly in the bedding plane. The shape of the anisotropy tensor is also fully considered in the revised correction by not constraining it to be oblate. Declination will also be affected by the correction, particularly if the ARM anisotropy tensor is triaxial.

6.2. Compaction Correction Results

[35] The compaction correction was applied to the ChRM of each individual sample using the revised correction equation and the corrected paleomagnetic directions were averaged to calculate the site means as in a standard paleomagnetic study. The corrected directions showed slightly better clustering at the site level showing an increase of k and decrease of α_{95} except for the UMC concretion site, which had a prolate fabric and the NG turbidite site, which had a strong triaxial fabric (Table 1). The difference of the clustering of the directions after the compaction correction was too small to be statistically significant at the 95% confidence level. The corrected site-mean inclinations became 4.3°–20.1° steeper than the original inclinations.

[36] The mean paleomagnetic direction for the Northumberland formation after the compaction correction using an *a* factor of 2.35 is $D = 6.1°$, $I = 60.5°$, $\alpha_{95} = 7.1°$, which

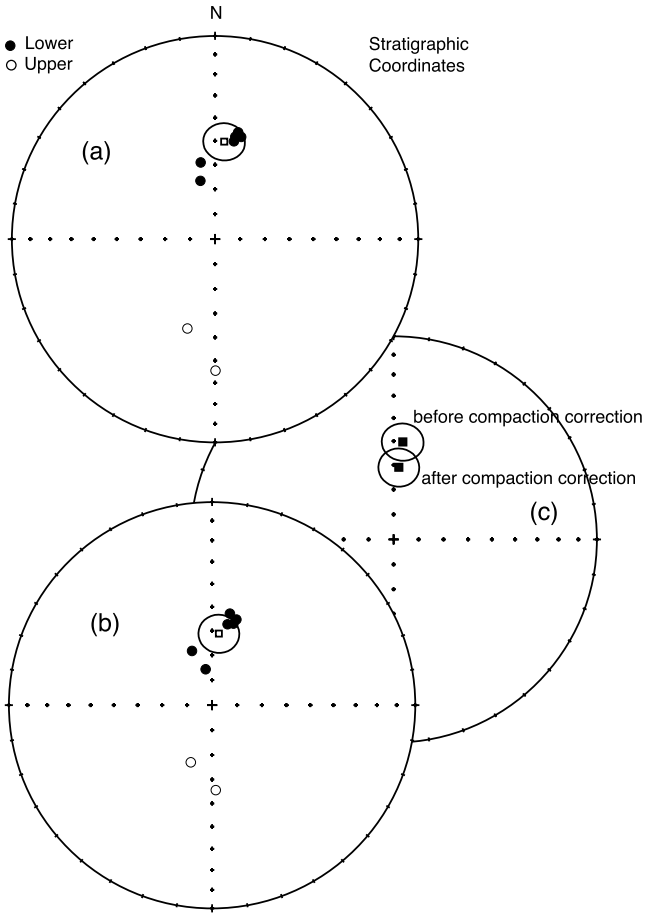


Figure 10. Equal-area projections of site mean directions (a) before and (b) after the compaction corrections. (c) The Northumberland formation mean directions before and after the compaction correction. These directions are statistically different at the 90% confidence level.

is 9.6° steeper than the mean inclination before the correction (Figure 10). When the mean directions before and after correction are compared in a fold test [*McFadden and Jones*, 1981], the corrected direction is significantly different from the original direction at the 90% confidence level. The a factors determined from the epoxy drying sample have a standard deviation of 0.55. If the range in a factors indicated by the standard deviation are used for the corrections ($a - 1$ SD = 1.80 and $a + 1$ SD = 2.90), the mean direction of the Northumberland formation ranges from D = 5.9°, I = 58.5°, α_{95} = 7.2° to D = 6.8°, I = 65.4°, α_{95} = 7.3°. These results show the corrected mean direction ranges from 7.6° to 14.5° steeper than the mean inclination before the correction and gives a sense of the error in the correction.

7. Discussion

7.1. Mean Paleomagnetic Direction

[37] The mean paleomagnetic direction from this study before correction is quite different from that of *Ward et al.* [1997] but statistically indistinguishable from that of *Enkin et al.* [2001] at the 95% confidence level. The mean direction of the Nanaimo Group by *Ward et al.* [1997] is 8°–13° shallower and rotated 24°–28° clockwise compared to the directions from *Enkin et al.* [2001] and this study (Figure 11). These paleomagnetic studies have differences in their sampling strategies, the statistical approaches used to calculate the site-mean directions, and the rejection rates of the samples used to calculate the mean directions. *Ward et al.* [1997] obtained the mean paleomagnetic direction of the Nanaimo Group by averaging 65 individually oriented samples from the Northumberland formation on Hornby Island and 38 samples from the Pender Formation on Texada Island. *Enkin et al.* [2001] sampled 67 sites (one site is the average of multiple individually oriented samples) from seven islands for the Nanaimo Group including 29 sites from Hornby Island. Only 17 sites were used to calculate the mean direction of the Nanaimo Group, and 9 sites of these were from Hornby Island. They could accept only 2 sites out of 4 collected from the Northumberland formation. We collected 36 sites from Hornby Island. Only 8 sites from the Northumberland formation were accepted for calculation of the formation mean. The rejection rate from Ward et al.'s study was only 27% (28 samples out of 103) while both *Enkin et al.* [2001] and our study showed high rejection rates of ~75% (50 sites out of 67 sites [*Enkin et al.*, 2001] and 28 sites out of 36 sites (our study)). The fundamental difference between our study and *Ward et al.*'s [1997] study is the way that Ward et al. define a site. They assume that the depositional rate of the Nanaimo sedimentary rocks is slow enough that one paleomagnetic core sample will average secular variation, and thus a standard paleomagnetic sample from the Nanaimo sedimentary rocks can be treated as a site (J. L. Kirschvink, personal communication, 2002). Since a sample is treated as a site, only the MAD can be used as a criterion to reject their paleomagnetic data, while Enkin et al.'s study and this study have an additional criterion that can be used to reject sites, the degree of scatter at a site based on its α_{95}.

[38] The lithology targeted by each study also differs from study to study. The paleomagnetic data of *Ward et al.* [1997] are from the mudstone and concretion samples from

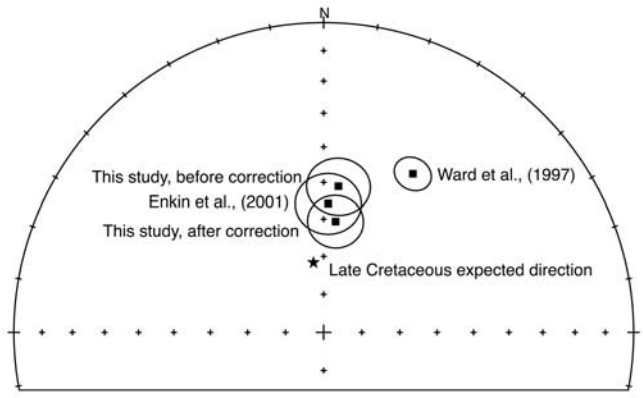


Figure 11. Equal-area projections of the paleomagnetic directions from the Nanaimo Group. The mean directions from this study and *Ward et al.* [1997] are from Hornby Island, while the mean direction from *Enkin et al.* [2001] is obtained from Hornby Island, Gabriola Island, and Mayne Island. The expected direction was calculated from a Late Cretaceous reference pole compiled by *Dickinson and Butler* [1998].

the Northumberland formation where they found ammonite and inoceramid bivalve fossils composed of unaltered aragonite. The presence of such unaltered fossil materials indicates that chemical or thermal remagnetization was minimal or absent. They argue that early diagenetic carbonate concretions are free from compaction effects, and statistically indistinguishable magnetic directions obtained from concretions and silty mudstones suggest no compaction effects in the silty mudstones. *Enkin et al.* [2001], however, focused more on the fine- to medium-grained sandstone because they believed that the mudstones were more susceptible to fractures, degradation of magnetites from high organic content, and shallowing of inclination by compaction.

[39] As shown in Figure 2, we collected mudstone, concretion and sandstone samples. The sandstones we collected in the younger formations above the Northumberland formation did not yield a stable remanence. *Enkin et al.*'s [2001] sandstone samples came from the southern shoreline of the island while our sandstone samples were collected along the northern shoreline, which might explain the difference in the results. While *Ward et al.* [1997] have not found much difference in directions between concretion and mudstone samples, our data suggest that mudstone samples have shallower inclinations than concretion samples. The reason why we see a difference between the mudstone and concretions is probably because our selection/rejection criteria were more stringent than in Ward et al.'s study.

7.2. Magnetic Mineralogy and Fabric

[40] The unblocking temperature near 580°C in thermal demagnetization of IRM and NRM suggests that the magnetic carriers of the ChRM are probably detrital magnetites and titanomagnetites with low Ti content. SEM observation also supports the presence of fine grained magnetites. In addition, SEM observation shows the presence of iron sulfides. Typically pyrite framboids were observed; therefore most of the iron sulfides in the samples are probably diagenetic pyrite. If pyrrhotite is present, it could explain the 400 °C unblocking temperature component, but pyrrhotite is not likely to contribute significantly to the remanence because the thermal demagnetization of IRM showed that magnetic minerals with low coercivity carried most of the remanence. The 200°C unblocking temperature component could be detrital titanomagnetite grains with the composition of $Fe_{2.4}Ti_{0.6}O_4$, which is typical for seafloor basalts.

[41] Most sites from our study showed depositional/compactional fabrics for both the AMS and AAR results. In many cases, the AMS had maximum and intermediate axes closer to the bedding plane than the AAR maximum and intermediate principal axes were. This observation suggests that there is a strong contribution of paramagnetic clay minerals to the AMS. The AMS maximum axes from the sites with strong triaxial fabric (UMC and NG) trend either north-south or east-west. These trends agree with the paleocurrent directions reported by *Katnick and Mustard* [2001]. Turbidite site (NG) showed an interesting triaxial fabric for both the AMS and the AAR results. While the AMS showed a composite fabric with a lineation imposed on the typical bedding parallel foliation, the AAR showed a distinct triaxial fabric with minimum axes tilted slightly from the vertical to the southeast. The AAR could reflect a current-affected depositional signal carried by small prolate ferromagnetic grains while a compactional fabric carried by all the ferromagnetic grains and the paramagnetic matrix, including the platy clay particles, dominated the AMS fabric.

[42] One of the significant observations of this study is that carbonate concretion samples carry a depositional-compactional AAR fabric with a magnetic anisotropy intensity up to ~20% (Figure 7). *Ward et al.* [1997] and *Wynne et al.* [1995] had both assumed that the paleomagnetic directions from mud rock sites should be free from compaction shallowing because the mud rock site means were indistinguishable from the concretion site means. However, this study demonstrates that concretion samples can also develop a depositional-compactional fabric, and their anisotropy requires a correction for the deflected paleomagnetic direction. This finding implies that the magnetic directions from the sedimentary rocks of Mount Tatlow [*Wynne et al.*, 1995] might also suffer from inclination shallowing and the northward translation of the Coast Mountains orogen proposed by Wynne et al. might have been overestimated.

7.3. Compaction Correction

[43] The compaction-corrected direction of the Northumberland formation has a significantly higher mean inclination (60.5°) than that reported by *Ward et al.* [1997] (42.5°). The mean inclination of the Nanaimo Group by *Enkin et al.* [2001] is slightly lower than this study's result (55.2°), but the two paleomagnetic directions are statistically indistinguishable at the 95% confidence level. The corrected inclination of 60.5° places the study area at a paleolatitude of 41.5°N at ~75 Ma. The Insular superterrane on which the Nanaimo Group was deposited has been argued to have ~3500 km of latitudinal displacement since the Late Cretaceous [*Ward et al.*, 1997], but the result of this study reduces the amount to 1600 km.

[44] The paleolatitude of 25°N from the Nanaimo Group [*Ward et al.*, 1997] would imply that Baja BC did not begin its northward translation until after 75 Ma, and the displacement rate would be 14 cm/yr in order to complete the translation by 50 Ma, which is faster than coast-parallel rates of 9–11 cm/yr calculated from a plate model assuming 100% coupling between the ocean plates and North America [*Harbert*, 1990]. The displacement trajectories of the Baja BC terranes constructed by *Housen and Beck* [1999] and *Kodama and Ward* [2001] based on *Debiche et al.*'s [1987] plate model places the 75 Ma Northumberland formation of the Nanaimo Group near 40°N, which agrees very well with this study's paleolatitude of 41°N from the corrected inclination (Figure 12). However, a relative motion history of Hawaiian hop spot [*Tarduno et al.*, 2003] suggests that the configuration of plates might have been different from the one constructed by *Debiche et al.* [1987], calling for a reexamination of the displacement trajectories of the Baja BC terranes. The change of hot spot reference could require more northerly location of Farallon and Kula plates, thus limiting the extent of driving forces for the latitudinal translation of Baja BC terranes. In order to have an accurate reconstruction of plate configuration, a relative motion history from hot spots remotely located from Hawaii is essential to account for true polar wonder, and the

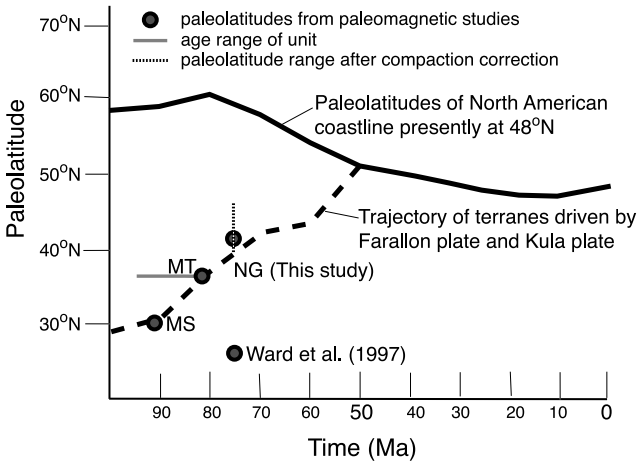


Figure 12. Paleolatitudes of North America and the Baja BC terranes constructed from the plate motion of the Farallon plate and Kula plate. Mount Stuart batholith (MS [*Housen et al.*, 2003]), Mount Tatlow (MT [*Wynne et al.*, 1995]), and the Nanaimo Group (NG (this study)) are located near 48°N, and their paleolatitudes are compared with the trajectory of Baja BC terrane. The paleolatitude from the Nanaimo Group (NG) after the inclination correction agrees with the expected paleolatitude from the construction. The trajectory of Baja BC terranes is shown as dashed line since the relative position of the Farallon and Kula plates to the North America plate is being questioned [*Tarduno et al.*, 2003]. Modified after *Kodama and Ward* [2001] and *Housen and Beck* [1999]. See color version of this figure in the HTML.

trajectory of Baja BC terranes should be reevaluated accordingly.

[45] *Kodama and Ward* [2001] estimated that in order to place the Nanaimo Group at the paleolatitude compatible with the distribution of rudists in the Late Cretaceous about 16° inclination shallowing would be required to correct the results of *Ward et al.* [1997]. However, our paleomagnetic direction, before the compaction correction, is already 8° steeper than that of *Ward et al.* [1997] and had to have suffered 9.6° inclination shallowing to yield the paleolatitude of 41°N. The amount of inclination shallowing from this study is close to the inclination shallowing reported from the fine-grained marine sedimentary rocks from coastal California (14° inclination shallowing for the Pigeon Point Formation, *Kodama and Davi* [1995]; 11°–17° inclination shallowing for the Ladd Formation and the Point Loma Formation, *Tan and Kodama* [1998]).

[46] The paleolatitude of the Nanaimo Group from this study is also supported by the geological evidence. *Housen and Beck* [1999] used the detrital zircons in the Nanaimo Group sedimentary rocks to show that the U-Pb ages of zircons reflected the change of the source area for the detrital zircons as the Nanaimo Group rocks moved northward. Fifteen degrees of latitudinal offset between the Nanaimo Group and cratonic North America based on our study would locate the area near the southwestern margin of North America at ~75 Ma as Housen and Beck suggested. The restored distribution of rudists using compaction-corrected paleolatitudes also suggests that the Nanaimo Group would have a displacement less than 1500 km in the Late Cretaceous [*Kodama and Ward*, 2001]. The distribution of rudists is confined to the paleolatitudes between 34°N and 40°N when the compaction-corrected paleolatitudes of the Cretaceous strata from the Peninsular Ranges and Salinia terranes that contain rudists are used. The absence of rudists in the Nanaimo Group sedimentary rocks indicates that the sedimentary rocks in the Nanaimo basin (Campanian-Maastrichtian) were deposited north of the paleolatitude of 40°N.

[47] The paleolatitude of the Nanaimo Group from this study also suggests that the Insular superterrane and the Intermontane superterrane have the same amount of displacement, with respect to their current North American positions, in the Late Cretaceous (~75 Ma) and there was not significant offset between these two terranes. This is well supported by geologic evidence as well as other paleomagnetic studies of Wrangellia terrane (Insular superterrane) and Intermontane superterrane rocks. The Okanogan Range batholith of the Intermontane superterrane was shown to be the source of the clasts found in the early Maastrichtian (71–68 Ma) Pipestone Canyon Formation, which overlies the Upper Jurassic Newby Group, the base of the southern Methow terrane [*Peterson et al.*, 1997]. This geological relationship means that the Insular and Intermontane superterranes had no significant offset since 71 Ma. The 70 Ma Carmacks Group from the Intermontane superterrane was revisited by *Wynne et al.* [1998] and their paleomagnetic data showed a displacement of 1900 ± 700 km. The samples were collected from subaerial volcanic rocks and passed the fold test and reversal test. The 80 Ma MacColl Ridge Formation in the Wrangellia terrane was revisited by *Stamatakos et al.* [2001], and they showed a displacement of 1650 ± 890 km, which is a significantly reduced amount compared to the previously suggested >4000 km [*Panuska*, 1985]. Their results are from volcanic and sedimentary rocks and showed better clustering in stratigraphic coordinates as well as reversed polarity directions.

[48] The paleomagnetic result from this study locates the Nanaimo Group and the Insular superterrane at a paleolatitude of 41°N at ~75 Ma, which agrees with the prediction from the modified Baja BC hypothesis by *Housen and Beck* [1999]. A moderate amount of displacement is suggested by the volcanic and sedimentary rocks of the Insular and Intermontane superterranes (the MacColl Ridge Formation (80 Ma [*Stamatakos et al.*, 2001]), the Northumberland formation (75 Ma (this study)), and the Carmacks Group (70 Ma [*Wynne et al.*, 1998]). The compaction-corrected latitudes of rudist distribution also suggest that the 80–75 Ma Nanaimo Group was situated north of 40°N [*Kodama and Ward*, 2001]. All of these results would suggest that the Baja BC terranes had 1600–1900 km of displacement for the time period of 80–70 Ma.

[49] *Haskin et al.*'s [2003] recent paleomagnetic study from the Empire Valley-Churn Creek area suggests that the Insular and Intermontane superterranes were linked by the mid-Cretaceous and that paleomagnetic data combined with the previous results of the Spences Bridge Group [*Irving et al.*, 1995] yield a southerly displacement of 1050 ± 450 km for the period of 105–100 Ma. However, *Enkin et al.* [2003] suggest that Upper Cretaceous strata of the Churn Creek area are equivalent facies of the Silverquick/Powell Creek succession from the Mount Tatlow area and that the combined

paleomagnetic data from Mount Tatlow and the Churn Creek areas indicate a southerly displacement of 3050 ± 450 km at 95–85 Ma. These results sharpen the conflict between the geologic evidence and paleomagnetic data and postulate a scenario of fast sourtherly and then northerly terrane motion in the Cretaceous ("yo-yo tectonics"). One possibility is that the data from Silverquick/Powell Creek succession of Mount Tatlow and Churn Creek areas suffer from inclination shallowing caused by burial compaction. In this paper, we present evidence that carbonate concretion samples could suffer from inclination shallowing. Hence the displacement of the Mount Tatlow strata might have been overestimated concretion data that is used to argue against inclination shallowing. *Enkin et al.* [2003] note that remanence anisotropy data of their samples from the Churn Creek area are inconsistent with inclination shallowing, but the anisotropy data they indicate are from only 10 samples. In order to fully evaluate the anisotropy fabric, it would be important to measure AAR of most of the samples that yielded the characteristic remanent directions. Our compaction correction for magnetite-bearing sedimentary rocks typically ranges from 10° to 20° of shallowing. If 10°–15° of inclination shallowing is removed from the data of Silverquick/Powell Creek succession, the latitudinal displacement would decrease from 3050 km to 1750–1000 km, and the complex motion history of the Baja BC terranes might not be required or at least the magnitude of southbound and northbound motion would be significantly reduced.

8. Conclusions

[50] The corrected inclination of the Northumberland formation locates the Nanaimo group at 41.5°N in the Late Cretaceous (~75 Ma). Oblate remanence fabrics indicate the presence of and allow the correction for compaction-caused inclination shallowing. Concretions do not always protect magnetic minerals from depositional-compactional inclination shallowing. The amount of inclination shallowing for DRM in magnetite appears to range from 10° to 20° for clay-rich sediments. The Baja BC hypothesis is constrained to indicate only about 1600 km offset of allochthonous terranes since 75 Ma. This result indicates that translation of large allochthonous terranes along active continental margins maybe an important tectonic process; however, the amount of offset is probably on the order of 1500 km or less, not the >3000 km previously proposed by the Baja BC hypothesis.

Appendix A: Determination of the *a* Factor Using Magnetic Grain Shape Ratios

[51] The relationship between normalized principal axes of AMS (p_i, $i = x, y, z$) and AAR (q_i, $i = x, y, z$) is shown [*Stephenson et al.*, 1986] to be

$$p_i = p_0 + sq_i \tag{A1}$$

$$p_0 = \frac{(a_r - a_m)}{(a_m + 2)(a_r - 1)} \tag{A2}$$

$$s = \left(\frac{a_m - 1}{a_m + 2}\right)\left(\frac{a_r + 2}{a_r - 1}\right) \tag{A3}$$

when a_m is the individual grain anisotropy of susceptibility and a_r is the individual grain anisotropy of remanence. For multidomain magnetite particles, *Stephenson et al.* [1986] have shown that

$$a_m = \frac{1 + N_n x_i}{1 + N_p x_i} \tag{A4}$$

$$a_r = \frac{N_n(1 + N_n x_i)}{N_p(1 + N_p x_i)} \tag{A5}$$

when N_n and N_p (= $1 - 2N_n$) are the demagnetizing factors normal and parallel to the revolution axis of the magnetic grain and χ_i is the intrinsic susceptibility. The grain shape ratio of the magnetic grains is used to derive the demagnetizing factors, N_n and N_p [*Stacey and Banerjee*, 1974, equation 3.8] and the principal axes of AMS and AAR are used to determine p_0 using the equation (A1). With p_0 and N_n thus determined, χ_i can be computed by combining the equations (A2), (A4), and (A5). Finally, with N_n and χ_i, the individual grain anisotropy factors, a_m or a_r can be calculated using equations (A4) and (A5).

Appendix B: Derivation of the Revised Compaction Correction Equation

[52] In estimating the DRM anisotropy tensor, the approach by *Tan and Kodama* [2003] is utilized. The ARM anisotropy tensor is calculated from the pARM measurements using nine different orientations, and the principal anisotropy axes in stratigraphic coordinates are obtained [*McCabe et al.*, 1985]. The relationship between the principal axes of bulk sample anisotropy (X_i, i = 1, 2, 3 for maximum, intermediate, and minimum principal axes) and the number densities (k_i, i = 1, 2, 3) of the individual particle's shape axes in *Stephenson*'s [1981] distribution function is shown by *Tan and Kodama* [2003]:

$$k_i = \frac{5(L + 2S)Xi - (L + 4S)}{2(L - S)} \tag{B1}$$

where L and S are the magnetic susceptibility parallel and perpendicular, respectively, to the axis of revolution of the individual magnetic particle. The principal axes of the ARM anisotropy tensor and DRM anisotropy tensor can be related to the number densities of the individual particle's principal axes by equation (B1). First, the principal axes of ARM anisotropy tensor are expressed as

$$X_{i\mathrm{ARM}} = \frac{2(a - 1)k_i + (a + 4)}{5(a + 2)} \tag{B2}$$

when prolate magnetic particle shape is assumed and the grain anisotropy, a, is defined by L/S. Second, the principal axes of the DRM anisotropy tensor have a simple form of equation (B1) when the depositional remanence is carried only along the unique, long axis, L of the magnetic grain:

$$k_i = \frac{X_{i\mathrm{DRM}} - 1}{2}. \tag{B3}$$

Finally, equating (B2) and (B3) derives the relationship between the principal axes of the ARM and DRM anisotropy tensors:

$$X_{i\mathrm{DRM}} = \frac{X_{i\mathrm{ARM}}(a+2)-1}{(a-1)} \tag{B4}$$

Equation (B4) indicates that once the principal axes of the ARM tensor and the individual grain anisotropy are known, the principal axes of the DRM tensor can be calculated. In constructing the DRM anisotropy tensor, the orientation of the DRM anisotropy tensor is assumed to be the same as the orientation of the ARM anisotropy tensor, however, it will be more anisotropic than the ARM tensor and this will be a function of the individual grain anisotropy. The eigenvectors of the DRM tensor are calculated from the orientations of the principal axes of the ARM tensor, $D_{i\mathrm{ARM}}$ and $I_{i\mathrm{ARM}}$, and the eigenvalues of the DRM tensor are the principal axes of the DRM anisotropy tensor ($X_{i\mathrm{DRM}}$) derived from the ARM anisotropy tensor by equation (B4). Since we are interested in restoring the original magnetic field direction (**H**), the inverse matrix of the DRM tensor is calculated and then multiplied by the characteristic detrital remanence direction (DRM) in stratigraphic coordinates to yield the original magnetic field vector components, H_x, H_y, and H_z, which are used to calculate the corrected inclination and declination.

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